

Global Volcano Activity Analysis: Patterns, Modeling, and Insights

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Problem Motivation

- Volcanic eruptions pose significant global risks, yet eruption records remain incomplete across regions and time.
- Missing and uneven VEI reporting limits reliable magnitude comparisons.
- A unified workflow is needed to clean data, visualize patterns, and apply predictive modeling.

Data & Study Design

Datasets Used: Smithsonian GVP Holocene Volcano List and GVP Eruption History Database, containing volcano attributes and eruption histories across ~12,000 years.

Integrated Study Dataset: Combined volcano-level features with eruption-level features, forming a unified dataset of ~10,000 eruptions.

Study Objective: Establish a structured dataset suitable for exploratory analysis and supervised learning tasks, including VEI prediction and dormancy classification.

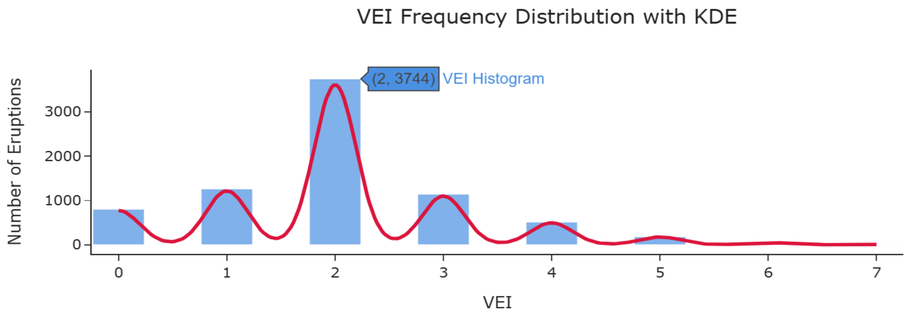
Data Processing Workflow

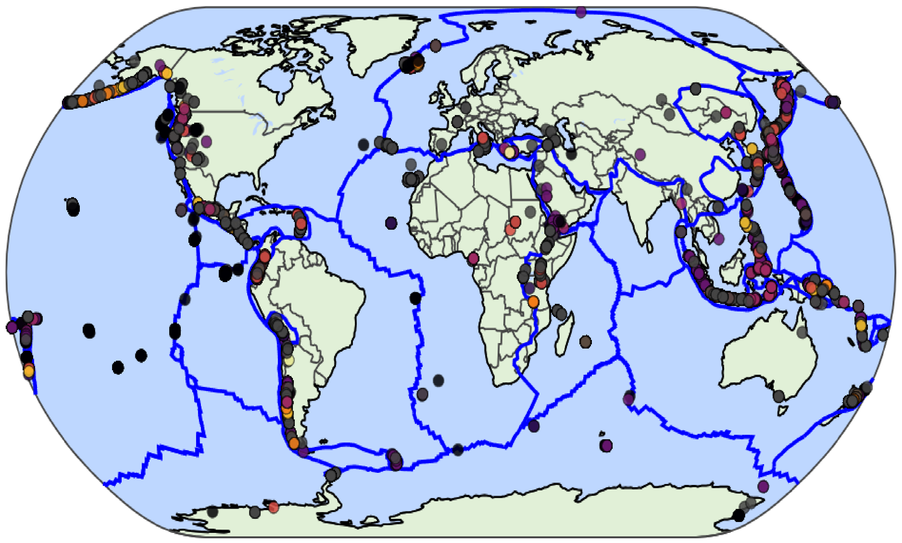
- Merged volcano and eruption datasets into a unified table.
- Removed irrelevant columns and handled missing values needed for EDA and dormancy calculation.
- Split the cleaned dataset into EDA and modeling tracks.
- Refined the modeling dataset through four updated versions for feature readiness and imputation.

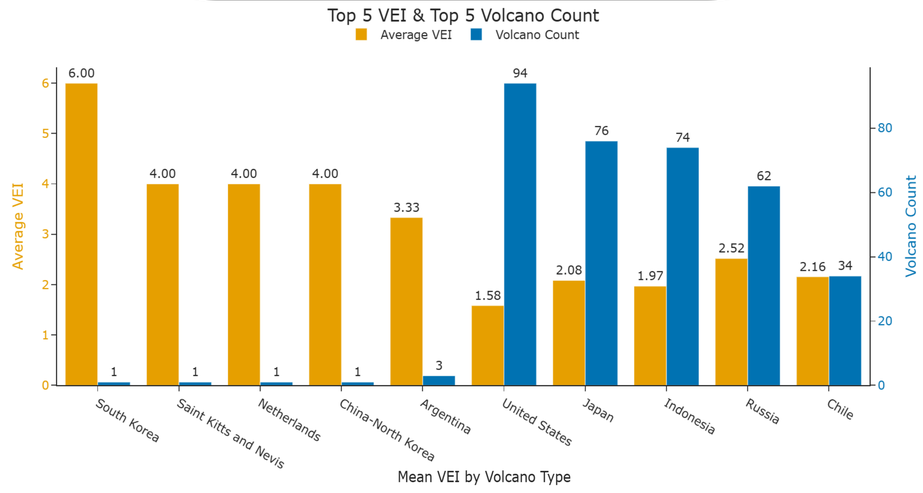
Methodology

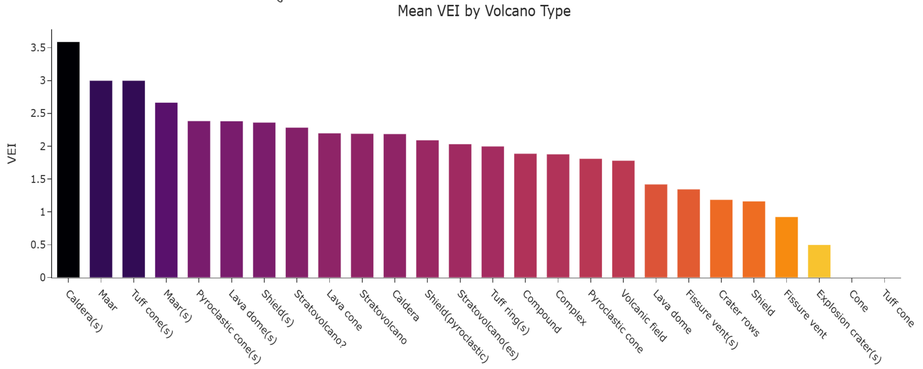
- Analyzed VEI trends, eruption frequency, volcano types, tectonics, and country activity.
- Mapped global volcano locations and tectonic boundaries.
- Created dormancy metrics and dormancy classes.
- Trained ML models for VEI imputation and dormancy classification.

Visual Analysis









Findings

- Elevation is not a reliable factor for VEI.
- Eruptions cluster mainly along convergent margins, especially the Pacific Ring of Fire.
- VEI depends strongly on tectonic setting and volcano type, with calderas and subduction zones showing higher magnitudes.
- Eruption activity increases toward the present.
- Rarely erupting volcanoes show higher VEI; frequently active ones show lower VEI.
- ML models identified clear dormancy groups and improved dataset completeness.

Limitations

- Historical eruption records are incomplete.
- Around 20% of eruptions lack VEI values.
- Recent increases in recorded eruptions may reflect reporting bias rather than true activity.
- Additional factors such as erupted volume, plume height, and duration are required for VEI estimation.

Future Work

- Integrate geophysical variables for improved modeling and enhanced predictive capability.
- Build advanced spatiotemporal eruption models.
- Engineer richer features for eruption dynamics.

References

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2. Aspinall, W. P., Woo, G., Voight, B., & Baxter, P. J. (2003). *Evidence-based volcanology: Application to eruption crises.* Journal of Volcanology and Geothermal Research, 128(1–3),273–285. https://doi.org/10.1016/S0377-0273(03)00260-9